

OMG Project - Networking Team Planning Document

Objectives

The Networking Team is responsible for:

- managing the multiplayer aspects of the game
 - Connection of both the players to the same game successfully
- handling real-time interactions
 - In game chatting
- server-client communication
 - Sends win/lose detail of a player after a game ends
- synchronization.
 - Saving/loading game
 - Game board state must be updated during player moves.
 - Allow players to reconnect if something goes wrong with timeout logic(give 5 sec to reconnect, if not then end game)

Implementation Approach

Technology Stack (To be probably taught in lectures)

- Networking Protocol: WebSockets (for real-time communication)
- Server Framework: Node.js with Socket.io or Python with FastAPI/WebSockets
- Client-Side Communication: WebSocket client integration
- Data Format: JSON for structured message exchange

Architecture Overview

- Client-Side:
 - Sends and receives game updates, chat messages, and connection status.

- Server-Side:
 - Manages active connections, processes game logic, and synchronizes game states.
- Database Integration (Stub Functions for Now):
 - Handles user authentication, matchmaking, and game history storage.

Things to Do for Iteration 1

Week 1

- Dummy Code Implementation
 - Create basic placeholder functions for networking logic.
 - Define communication stubs for server-client interaction. (If not possible then week 2)
 - Structure introductory networking architecture for better understanding.
- Step 1: Use Case Diagram
 - Design and finalize a use case diagram representing core networking functionalities.
 - Ensure alignment with matchmaking, game state synchronization, and in-game communication.
- Step 2: Use Case Descriptions
 - Write detailed use case descriptions for key networking functionalities based on the description.

Week 2

- Class Structure
 - Define object-oriented class structures
 - Develop sequence diagrams outlining real-time interactions and data flow.
- Further Completion of Dummy Code
 - Expand the initial dummy code with additional placeholders for networking functions.
 - Introduce method stubs for handling game states, chat functionality, and connection recovery.

